APPENDIX- A

SOURCE CODE

ProfileActivity.java

```
package com.example.biometricattendance;
import android.Manifest;
import android.content.Intent;
import android.content.pm.PackageManager;
import android.location.Location;
import android.location.LocationManager;
import android.os.Bundle;
import android.provider.Settings;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.biometric.BiometricPrompt;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
import java.text.SimpleDateFormat;
import java.util.Date;
import java.util.Locale;
import java.util.concurrent.Executor;
public class ProfileActivity extends AppCompatActivity {
    TextView emailText, attendanceText;
    Button markAttendanceBtn, viewRecordsBtn, logoutBtn;
    DatabaseHelper dbHelper;
    String userEmail;
    // Predefined allowed location (e.g., office/school)
    private static final double ALLOWED_LAT = 12.9716;
```

```
    private static final double ALLOWED_LNG = 77.5946;
    private static final float LOCATION_TOLERANCE_METERS = 100f; // 100
meters

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_profile);
        emailText = findViewById(R.id.profile_email);
        attendanceText = findViewById(R.id.attendance_count);
        markAttendanceBtn = findViewById(R.id.mark_attendance_btn);
        viewRecordsBtn = findViewById(R.id.view_records_btn);
        logoutBtn = findViewById(R.id.logout_btn);
        dbHelper = new DatabaseHelper(this);
        userEmail = getIntent().getStringExtra("email");
        emailText.setText("Welcome: " + userEmail);
        updateAttendanceCount();
        markAttendanceBtn.setOnClickListener(v ->
checkLocationAndMarkAttendance());
        viewRecordsBtn.setOnClickListener(v -> {
            Intent intent = new Intent(ProfileActivity.this,
AttendanceRecordsActivity.class);
            intent.putExtra("email", userEmail);
            startActivity(intent);
        });
        logoutBtn.setOnClickListener(v -> {
            Intent intent = new Intent(ProfileActivity.this, LoginActivity.class);
            startActivity(intent);
            finish();
        });
    }
```

```
    private void updateAttendanceCount() {
        int count = dbHelper.getAttendanceCount(userEmail);
        attendanceText.setText("Total Present Days: " + count);
    }

    private void checkLocationAndMarkAttendance() {
        LocationManager locationManager = (LocationManager)
getSystemService(LOCATION_SERVICE);
        boolean isGPSEnabled =
locationManager.isProviderEnabled(LocationManager.GPS_PROVIDER);
        boolean isNetworkEnabled =
locationManager.isProviderEnabled(LocationManager.NETWORK_PROVIDER);
        if (!isGPSEnabled && !isNetworkEnabled) {
            Toast.makeText(this, "Location services are disabled. Please turn on GPS.",
Toast.LENGTH_LONG).show();
           startActivity(new
Intent(Settings.ACTION_LOCATION_SOURCE_SETTINGS));
            return;
        }
        Location location = null;
        if (ActivityCompat.checkSelfPermission(this,
Manifest.permission.ACCESS_FINE_LOCATION) ==
PackageManager.PERMISSION_GRANTED) {
            if (isGPSEnabled)
                location =
locationManager.getLastKnownLocation(LocationManager.GPS_PROVIDER);
            if (location == null && isNetworkEnabled)
                location =
locationManager.getLastKnownLocation(LocationManager.NETWORK_PROVID
ER);
        } else {
```

```
            Toast.makeText(this, "Location permission not granted",
Toast.LENGTH_SHORT).show();
            return;
        }
        if (location != null) {
            double currentLat = location.getLatitude();
            double currentLng = location.getLongitude();
            float[] distance = new float[1];
            Location.distanceBetween(currentLat, currentLng, ALLOWED_LAT,
ALLOWED_LNG, distance);
            if (distance[0] < LOCATION_TOLERANCE_METERS) {
                authenticateAndMark("Present");
            } else {
                markAttendance("absent");
                Toast.makeText(this, "marked as absent",
Toast.LENGTH_SHORT).show();
                updateAttendanceCount();
            }
        } else {
            Toast.makeText(this, "unable to get the location",
Toast.LENGTH_SHORT).show();
        }
    }
    private void authenticateAndMark(String statusToMark) {
        Executor executor = ContextCompat.getMainExecutor(this);
        BiometricPrompt biometricPrompt = new
BiometricPrompt(ProfileActivity.this, executor,
                new BiometricPrompt.AuthenticationCallback() {
                    @Override
                    public void onAuthenticationSucceeded(@NonNull
BiometricPrompt.AuthenticationResult result) {
```

```
                super.onAuthenticationSucceeded(result);
                markAttendance(statusToMark);
                Toast.makeText(ProfileActivity.this, "Attendance marked: " +
statusToMark, Toast.LENGTH_SHORT).show();
                updateAttendanceCount();
            }
            @Override
            public void onAuthenticationFailed() {
                super.onAuthenticationFailed();
                Toast.makeText(ProfileActivity.this, "Fingerprint not recognized",
Toast.LENGTH_SHORT).show();
            }
        });
    BiometricPrompt.PromptInfo promptInfo = new
BiometricPrompt.PromptInfo.Builder()
        .setTitle("Confirm Attendance")
        .setSubtitle("Scan fingerprint to mark " + statusToMark)
        .setNegativeButtonText("Cancel")
        .build();
    biometricPrompt.authenticate(promptInfo);
  }
  private void markAttendance(String status) {
    String date = new SimpleDateFormat("yyyy-MM-dd",
Locale.getDefault()).format(new Date());
    String time = new SimpleDateFormat("HH:mm:ss",
Locale.getDefault()).format(new Date());
    dbHelper.markAttendance(userEmail, date, time, status);
  }
}
```

activity_profile.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/profile_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp"
    android:gravity="center"
    android:background="#FFFFFF">
    <TextView
        android:id="@+id/profile_email
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome: user@example.com"
        android:textSize="18sp"
        android:textStyle="bold"
        android:layout_marginBottom="16dp" />
    <TextView
        android:id="@+id/attendance_count"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Total Present Days: 0"
        android:textSize="16sp"
        android:layout_marginBottom="32dp" />
    <Button
        android:id="@+id/mark_attendance_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Mark Attendance"
```

```
        android:layout_marginBottom="16dp"
        android:backgroundTint="#4CAF50"
        android:textColor="#FFFFFF" />
    <Button
        android:id="@+id/view_records_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="View Attendance Records"
        android:layout_marginBottom="16dp"
        android:backgroundTint="#2196F3"
        android:textColor="#FFFFFF" />

    <Button
        android:id="@+id/logout_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Logout"
        android:backgroundTint="#F44336"
        android:textColor="#FFFFFF" />
</LinearLayout>
```

RegisterActivity.java

```
// RegisterActivity.java
package com.example.biometricattendance;
import android.annotation.SuppressLint;
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
```

```
import androidx.appcompat.app.AppCompatActivity;
import androidx.biometric.BiometricPrompt;
import androidx.core.content.ContextCompat;
import java.util.concurrent.Executor;
public class RegisterActivity extends AppCompatActivity {
    EditText etEmail, etPassword;
    Button btnRegister;
    DatabaseHelper databaseHelper;
    BiometricPrompt biometricPrompt;
    BiometricPrompt.PromptInfo promptInfo;
    String email = "";
    String password = "";
    String generatedFingerprint = "";
    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_register);
       etEmail = findViewById(R.id.et_register_email);
        etPassword = findViewById(R.id.et_register_password);
        btnRegister = findViewById(R.id.btn_register);
        databaseHelper = new DatabaseHelper(this);
        setupBiometricPrompt();
        btnRegister.setOnClickListener(v -> {
            email = etEmail.getText().toString().trim();
            password = etPassword.getText().toString().trim();
            if (email.isEmpty() || password.isEmpty()) {
                Toast.makeText(this, "Fill all fields", Toast.LENGTH_SHORT).show();
                return;
            }
            if (databaseHelper.isEmailRegistered(email)) {
```

```
                Toast.makeText(this, "Email already registered",
Toast.LENGTH_SHORT).show();
                return;
            }
            biometricPrompt.authenticate(promptInfo);
        });
    }
    private void setupBiometricPrompt() {
        Executor executor = ContextCompat.getMainExecutor(this);
        biometricPrompt = new BiometricPrompt(this, executor, new
BiometricPrompt.AuthenticationCallback() {
            @Override
        public void
onAuthenticationSucceeded(BiometricPrompt.AuthenticationResult result) {
                super.onAuthenticationSucceeded(result);
                generatedFingerprint = email + "_fingerprint"; // Unique fingerprint
binding
                boolean success = databaseHelper.registerUser(email, password,
 generatedFingerprint);
                if (success) {
                    Toast.makeText(RegisterActivity.this, "Registration Successful",
Toast.LENGTH_SHORT).show();
                    startActivity(new Intent(RegisterActivity.this, LoginActivity.class));
                    finish();
                } else {
                    Toast.makeText(RegisterActivity.this, "Registration Failed",
Toast.LENGTH_SHORT).show();
                }
            }

            @Override
```

```
            public void onAuthenticationFailed() {
                super.onAuthenticationFailed();
                Toast.makeText(RegisterActivity.this, "Fingerprint not recognized",
Toast.LENGTH_SHORT).show();
            }
        });
        promptInfo = new BiometricPrompt.PromptInfo.Builder()
                .setTitle("Fingerprint Registration")
                .setSubtitle("Scan your fingerprint to register")
                .setNegativeButtonText("Cancel")
                .build();
    }
}
```

activity_register.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">
    <EditText
        android:id="@+id/et_register_email"
        android:layout_width="match_parent"
        android:layout_height="48dp"
        android:hint="Enter Email"
        android:inputType="textEmailAddress"
        android:padding="10dp" />
    <EditText
        android:id="@+id/et_register_password"
```

```
        android:layout_width="match_parent"
        android:layout_height="48dp"
        android:hint="Enter Password"
        android:inputType="textPassword"
        android:padding="10dp"
        android:layout_marginTop="16dp" />
    <Button
        android:id="@+id/btn_register"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Register"
        android:layout_marginTop="24dp" />
</LinearLayout>
```

LoginActivity.java

```
package com.example.biometricattendance;
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.biometric.BiometricPrompt;
import androidx.core.content.ContextCompat;
import java.util.concurrent.Executor;
public class LoginActivity extends AppCompatActivity {
    EditText emailInput, passwordInput;
    Button loginBtn;
    DatabaseHelper dbHelper;
    String enteredEmail;
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_login);
        emailInput = findViewById(R.id.login_email);
        passwordInput = findViewById(R.id.login_password);
        loginBtn = findViewById(R.id.login_button);
        dbHelper = new DatabaseHelper(this);
        loginBtn.setOnClickListener(view -> {
            enteredEmail = emailInput.getText().toString();
            String password = passwordInput.getText().toString();
            if (dbHelper.checkUser(enteredEmail, password)) {
                authenticateFingerprint();
            } else {
                Toast.makeText(LoginActivity.this, "Invalid email or password",
Toast.LENGTH_SHORT).show();
            }
        });
    }
    private void authenticateFingerprint() {
        Executor executor = ContextCompat.getMainExecutor(this)
        BiometricPrompt biometricPrompt = new
BiometricPrompt(LoginActivity.this, executor,
                new BiometricPrompt.AuthenticationCallback() {
                    @Override
                    public void
onAuthenticationSucceeded(BiometricPrompt.AuthenticationResult result) {
                        super.onAuthenticationSucceeded(result);
                        Toast.makeText(LoginActivity.this, "Fingerprint Verified!",
Toast.LENGTH_SHORT).show();
```

```
                Intent intent = new Intent(LoginActivity.this, ProfileActivity.class);
                intent.putExtra("email", enteredEmail);
                startActivity(intent);
                finish();
            }
            @Override
            public void onAuthenticationFailed() {
                super.onAuthenticationFailed();
                Toast.makeText(LoginActivity.this, "Fingerprint not recognized",
Toast.LENGTH_SHORT).show();
            }
        });
    BiometricPrompt.PromptInfo promptInfo = new
BiometricPrompt.PromptInfo.Builder()
        .setTitle("Login Authentication")
        .setSubtitle("Scan your fingerprint to login")
        .setNegativeButtonText("Cancel")
        .build()
    biometricPrompt.authenticate(promptInfo);
  }
}
```

Activity_Login.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:gravity="center"
  android:padding="24dp">
  <TextView
```

```
        android:text="Login"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="24dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" />
    <EditText
        android:id="@+id/login_email"
        android:hint="Email"
        android:inputType="textEmailAddress"
        android:layout_width="match_parent"
        android:layout_height="48dp"
        android:layout_marginBottom="16dp"/>
    <EditText
        android:id="@+id/login_password"
        android:hint="Password"
        android:inputType="textPassword"
        android:layout_width="match_parent"
        android:layout_height="48dp"
        android:layout_marginBottom="24dp"/>
    <Button
        android:id="@+id/login_button"
        android:text="Login"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />
</LinearLayout>
```

AttendanceRecordsActivity.java

```
package com.example.biometricattendance;
import android.database.Cursor;
```

```
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import java.util.ArrayList;
public class AttendanceRecordsActivity extends AppCompatActivity {
    ListView recordsListView;
    DatabaseHelper dbHelper;
    String currentEmail;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_attendance_records);
        recordsListView = findViewById(R.id.recordsListView);
        dbHelper = new DatabaseHelper(this);
        currentEmail = getIntent().getStringExtra("email");
        if (currentEmail == null) {
            Toast.makeText(this, "No user email provided.",
Toast.LENGTH_SHORT).show();
            return;
        }
        loadAttendanceRecords();
    }
    private void loadAttendanceRecords() {
        Cursor cursor = dbHelper.getAllAttendanceRecords(currentEmail);
        ArrayList<String> records = new ArrayList<>();
        if (cursor != null && cursor.moveToFirst()) {
            do {
                String date = cursor.getString(cursor.getColumnIndexOrThrow("date"));
                String time = cursor.getString(cursor.getColumnIndexOrThrow("time"));
```

```
            String status =
cursor.getString(cursor.getColumnIndexOrThrow("status"));
            records.add(date + " " + time + " - " + status);
        } while (cursor.moveToNext());
        cursor.close();
    } else {
        records.add("No attendance records found.");
    }
    ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
android.R.layout.simple_list_item_1, records);
    recordsListView.setAdapter(adapter);
  }
}
```

activity_attendance_records.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="16dp">
<ListView
    android:id="@+id/recordsListView"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:dividerHeight="1dp"/>
</LinearLayout>
```

DatabaseHelper.java

```
package com.example.biometricattendance;
import android.content.ContentValues;
```

```
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
public class DatabaseHelper extends SQLiteOpenHelper {
    private static final String DATABASE_NAME = "AttendanceDB.db";
    private static final int DATABASE_VERSION = 1;
    // Users Table
    private static final String TABLE_USERS = "users";
    private static final String COLUMN_EMAIL = "email";
    private static final String COLUMN_PASSWORD = "password";
    private static final String COLUMN_FINGERPRINT = "fingerprint_data"; //
stub column for uniqueness
    private static final String TABLE_ATTENDANCE = "attendance";
    private static final String COLUMN_ID = "id";
    private static final String COLUMN_DATE = "date";
    private static final String COLUMN_TIME = "time";
    private static final String COLUMN_STATUS = "status";
    public DatabaseHelper(Context context) {
        super(context, DATABASE_NAME, null, DATABASE_VERSION);
    }
    @Override
    public void onCreate(SQLiteDatabase db) {
        String createUsersTable = "CREATE TABLE " + TABLE_USERS + " (" +
                COLUMN_EMAIL + " TEXT PRIMARY KEY, " +
                COLUMN_PASSWORD + " TEXT, " +
                COLUMN_FINGERPRINT + " TEXT UNIQUE)";
        String createAttendanceTable = "CREATE TABLE " +
TABLE_ATTENDANCE + " (" +
                COLUMN_ID + " INTEGER PRIMARY KEY AUTOINCREMENT, "
+
```

```
            COLUMN_EMAIL + " TEXT, " +
            COLUMN_DATE + " TEXT, " +
            COLUMN_TIME + " TEXT, " +
            COLUMN_STATUS + " TEXT)";
    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_USERS);
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_ATTENDANCE);
        onCreate(db);
    }
    public boolean registerUser(String email, String password, String
fingerprintData) {
        if (isFingerprintRegistered(fingerprintData)) return false;
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues values = new ContentValues();
        values.put(COLUMN_EMAIL, email);
        values.put(COLUMN_PASSWORD, password);
        values.put(COLUMN_FINGERPRINT, fingerprintData);
        long result = db.insert(TABLE_USERS, null, values);
        return result != -1;
    }
    public boolean checkUser(String email, String password) {
        SQLiteDatabase db = this.getReadableDatabase();
        Cursor cursor = db.query(TABLE_USERS, null, COLUMN_EMAIL + "=?
AND " + COLUMN_PASSWORD + "=?",
            new String[]{email, password}, null, null, null);
        boolean exists = (cursor != null && cursor.moveToFirst());
        if (cursor != null) cursor.close();
        return exists;
    }
    public boolean isFingerprintRegistered(String fingerprintData) {
```

```
        SQLiteDatabase db = this.getReadableDatabase();
        Cursor cursor = db.query(TABLE_USERS, null, COLUMN_FINGERPRINT
+ "=?",
                new String[]{fingerprintData}, null, null, null);
        boolean exists = (cursor != null && cursor.moveToFirst());
        if (cursor != null) cursor.close();
        return exists;
    }
    public boolean markAttendance(String email, String date, String time, String
status) {
        // Check if already marked for the date
        SQLiteDatabase db = this.getReadableDatabase();
        Cursor cursor = db.query(TABLE_ATTENDANCE, null,
                COLUMN_EMAIL + "=? AND " + COLUMN_DATE + "=?",
                new String[]{email, date}, null, null, null);
        if (cursor != null && cursor.moveToFirst()) {
            cursor.close();
            return false; // Already marked
        }
        ContentValues values = new ContentValues();
        values.put(COLUMN_EMAIL, email);
        values.put(COLUMN_DATE, date);
        values.put(COLUMN_TIME, time);
        values.put(COLUMN_STATUS, status);
        long result = db.insert(TABLE_ATTENDANCE, null, values);
        return result != -1;
    }
    public int getAttendanceCount(String email) {
        SQLiteDatabase db = this.getReadableDatabase();
        Cursor cursor = db.rawQuery("SELECT COUNT(*) FROM " +
TABLE_ATTENDANCE +
```

```
            " WHERE " + COLUMN_EMAIL + "=? AND " + COLUMN_STATUS
+ "=?", new String[]{email, "Present"});
        int count = 0;
        if (cursor.moveToFirst()) {
            count = cursor.getInt(0);
        }
        cursor.close();
        return count;
    }
public Cursor getAttendanceRecords(String email) {
        SQLiteDatabase db = this.getReadableDatabase();
        return db.query(TABLE_ATTENDANCE, null, COLUMN_EMAIL + "=?",
                new String[]{email}, null, null, COLUMN_DATE + " DESC");
    }
    public boolean isEmailRegistered(String email) {
        SQLiteDatabase db = this.getReadableDatabase();
        Cursor cursor = db.query("users", null, "email=?", new String[]{email}, null,
null, null);
        boolean exists = (cursor != null && cursor.moveToFirst());
        if (cursor != null) {
            cursor.close();
        }
        return exists;
    }
    public Cursor getAllAttendanceRecords(String email) {
        SQLiteDatabase db = this.getReadableDatabase();
        return db.query("attendance", null, "email=?", new String[]{email}, null, null,
"date DESC");
    }
}
```

APPENDIX- B

SCREENSHOTS

Sample Output

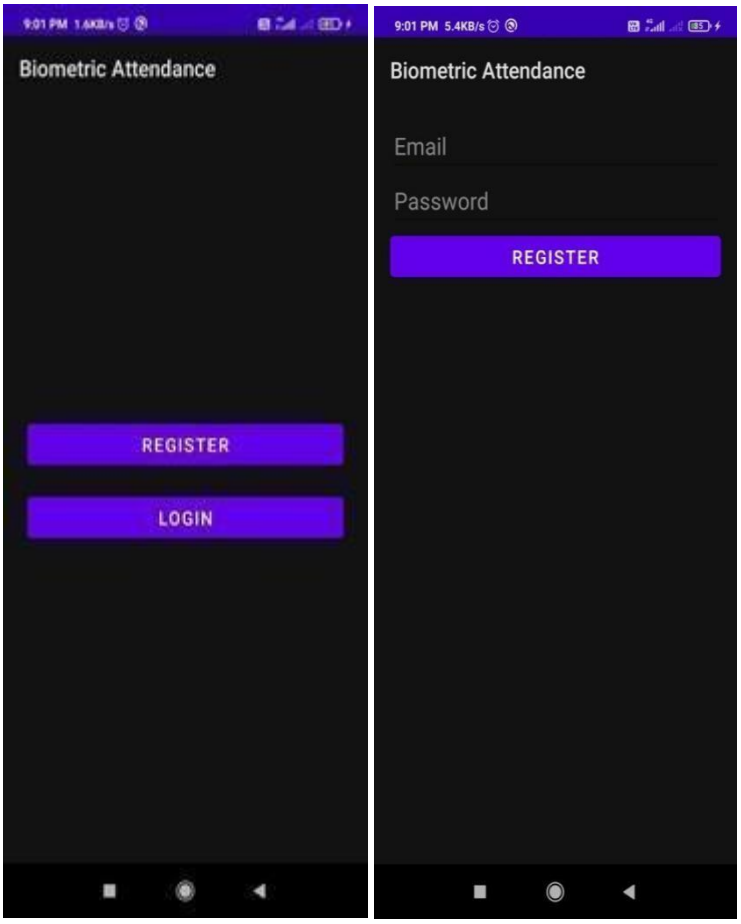

Figure B.1: User Registration

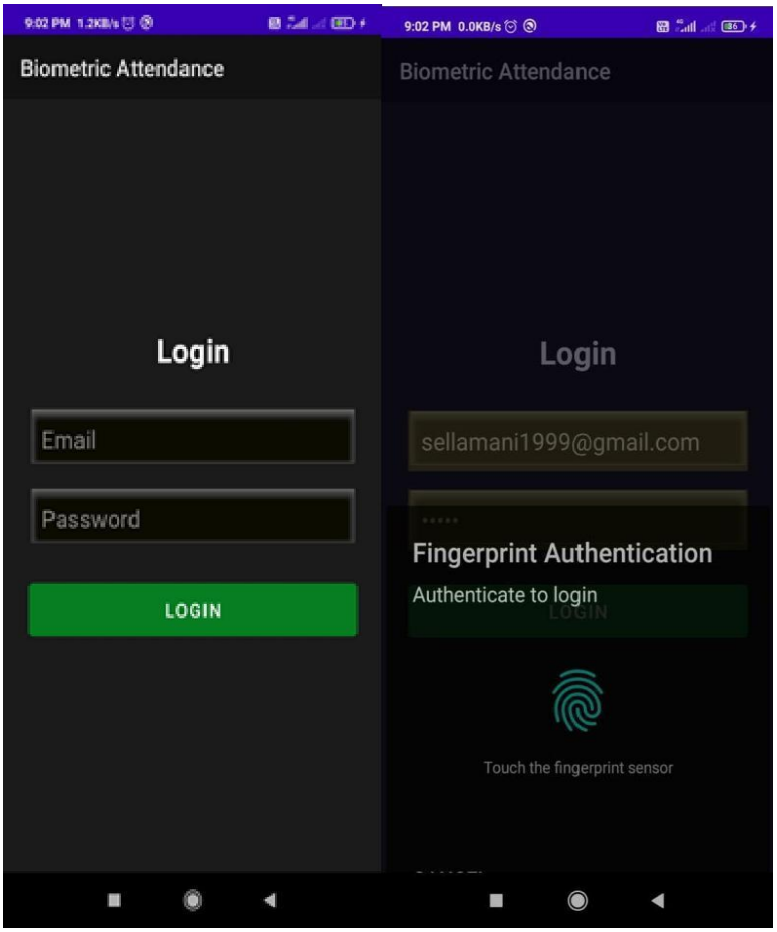

Figure B.2:User Login & Fingerprint Authentication

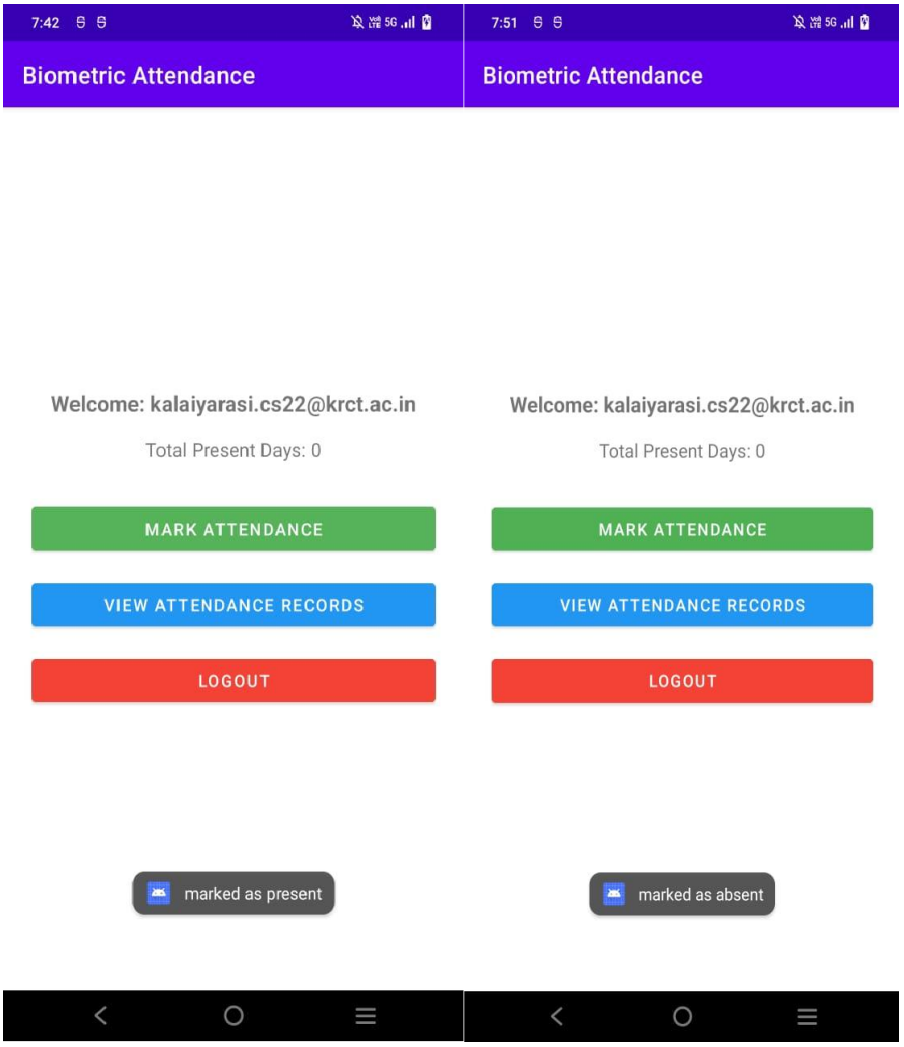

Figure B.3:Attendance marking

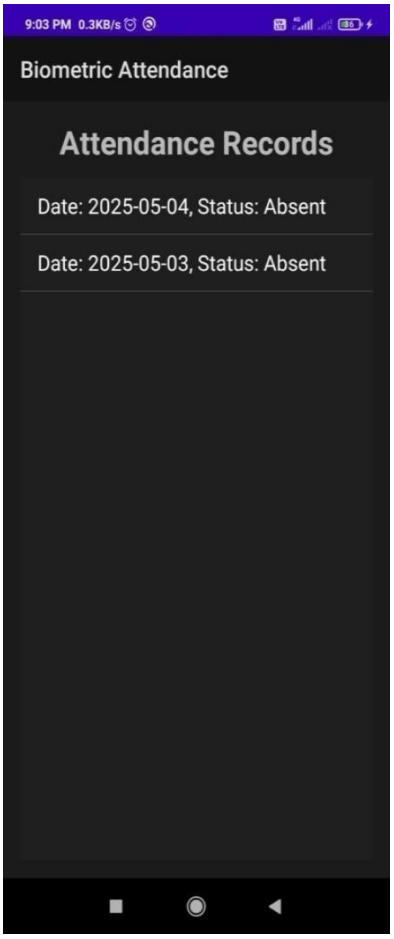

Figure B.4:Attendance Record